

Product overview

The relay has improved conductivity and heat dissipation capabilities on key components that can be used in thousands of higher power and temperature environments, such as charging applications. The PR60 two-pole power relay is ideal for charging, HVAC, industrial control, power and lighting applications.

Features&benefits

- ◇DPST-NO DPST-NC and DPDT configuration
- ◇Meets 8mm creepage, 4kV dielectric
- ◇Epoxy sealed version available
- ◇UL Class F (155°C) standard

Product certification

c UL us	UL	UL 60947-4-1	RoHS	RoHS	RoHS Compliant

Product Model

	PR60	2A	24	D	E	F

Type

Contact Form
2A：2 groups normally open
2B：2 groups normally close
2C：2 groups conversion

Coil Voltage:DC:5/6/12/24/48/110
AC:12/24/120/220/240/277

Voltage specification:
D: DC A5: AC 50Hz A6: AC 50/60Hz

Protection regulation
Nil: No seals E: seal

pin specification
F: 6.35×0.8mm KF: 4.75×0.5mm PF: PCB

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General Data

Life Expectancy Mechanical Electrical	Minimum operations 5×10^6 1×10^5 at 30A, 277VAC Res. (N.0.)
Operate Time	15ms typical 25ms maximum with bounce
Release Time	10ms typical 25ms maximum with bounce (with no coil suppression)
Dielectric Strength (at sea level for 1 min.)	1500Vrms contact to contact 4000Vrms contact to coil 2000Vrms between contact sets
Insulation Resistance	1000MΩ (500VDC)
Dropout	DC:Greater than 10%of nominal coil voltage AC: Greater than 20%of nominal coil voltage
Ambiant Temperature Operating	At nominal coil voltage DC: -40℃ (-40°F) to 85℃ (185°F) AC: -40℃ (-40°F) to 65℃ (149°F) -40℃ (-40°F) to 105℃ (221°F)
Vibration	0.062" (1.65mm) DA at 10-55Hz
Shock	Operational, 10g for 11ms 1/2 sine pulse Wno contact opening > 100usec) Non-destructive, 100g for 11ms 1/2 sine pulse
Enclosure	P.B.T. polyester
Terminals	Quick connect tabs Note: Allow suitable slack on leads when wiring. and do not subject the terminals to excessive force.
Max. Immersion Time	30 seconds
Weight	86 grams
Packing unit in pos	20 per plastic tray / 100 per carton box

Contacts

Arrangement	DPST(2 Form A) DPST(2 FormB) DPDT(2 Form C)
Ratings	Resistive load: Max. switched power: 1200 W or 11080 VA Max. switched current: 40A (N.O), 3A (N.C.) Max.switchedvoltage:30VDC*or600VAC *Notertf switching voltage is greater than 30VDC, special precautions must be taken. Please contact the factory.
Rated Load UL	Normally open contacts (N.O.) 40A at 277VAC, Resistive, 30k cycles [2][3] 30A at 277VAC, General Use, 100k cycles [2][3] 10A at 600VAC, General Use, 6k cycles [1][3] 1HP at 120VAC, 50k cycles [1][3] 2HP at 240VAC, 50k cycles [1][3] 2HP at 600VAC, 100k cycles [1][3] Normally open contacts (N.O.), DC Coils only 25.3FLA/ 110LRA at 240VAC, 100k cycles [1][3] Normally closed contacts (N.C.) 3A at 277VAC, General Use, 100k cycles [2][3] 3A at 250VAC, General Use, 100k cycles [2][3] 3A at 240VAC, General Use, 100k cycles [2][3] 3A at 125VAC, General Use, 100k cycles [2][3] 3FLA/ 3LRA at 240VAC, 30k cycles [1][3]
Resistance	≤50mΩ（@24V/1A)

Notes

1. All values at 40°C (104°F).
2. All values at 85°C (185°F).
3. Relay may pull in with less than “Must Operate” value.
4. Specifications subject to change without notice.

Coil Specifications

DC Coil

Nominal Coil VDC	Must Operate VDC	Max. Continuous VDC	Nominal Current mA ± 10%	Coil Resistance Ohm ± 10%
5	3.8	8.0	326.7	15.3
6	4.5	10.5	272.0	22
12	9.0	20.7	140.0	86
24	18.0	41.8	68.5	350
48	36.0	83.4	34.5	1390
110	82.5	190.5	15.2	7255

AC Coil

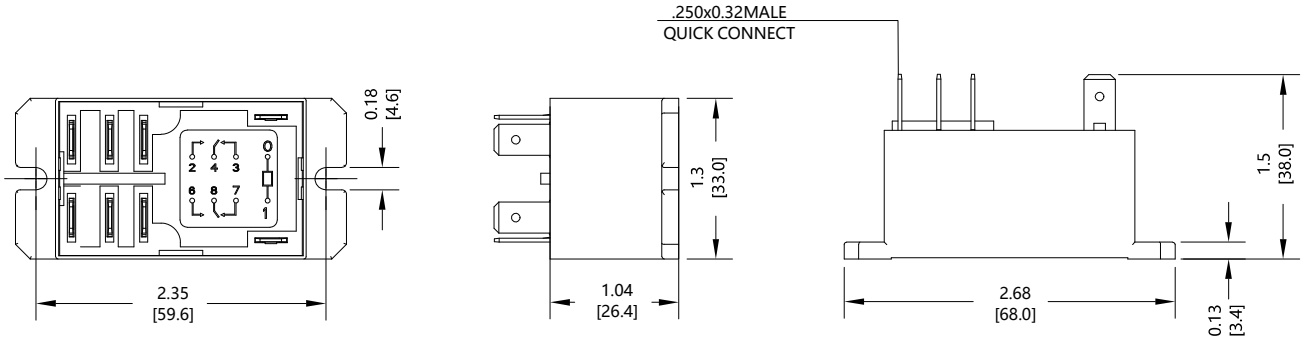
Nominal Coil VAC	Must Operate VAC	Max. Continuous VAC	Nominal Current mA ± 10%	50Hz Coil Resistance Ohm ± 10%	60Hz Coil Resistance Ohm ± 10%
12	9.6	15.6	340.0	9.5	8
24	19.2	31.2	166.0	45	35.7
120	96.0	156.0	33.3	1125	830
220	176.0	286.0	18.2	3800	2870
240	192.0	312.0	16.7	4500	3800
277	221.6	360.1	14.4	5960	4700

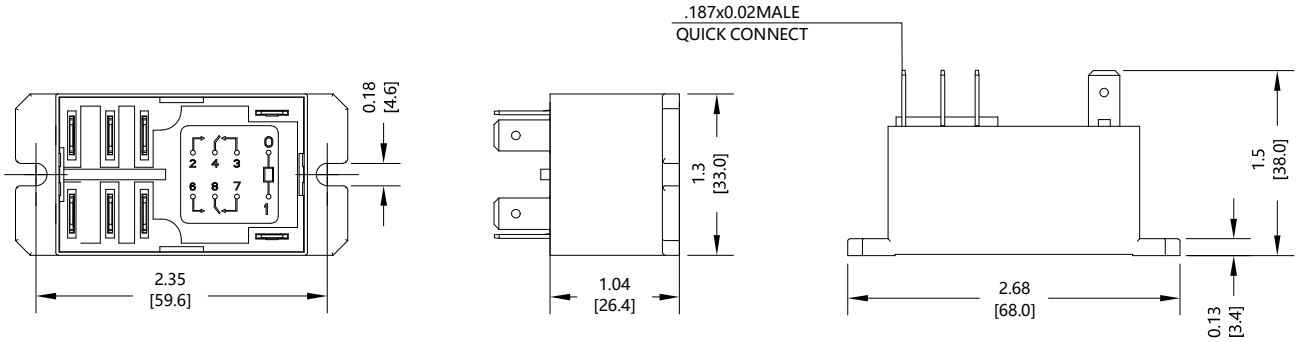
Power consumption

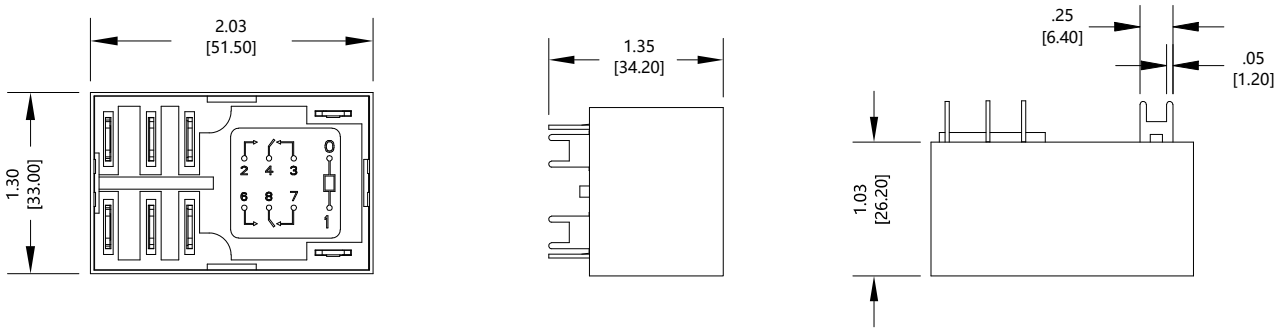
At Pickup Voltage	925mW DC coil
At Pickup Voltage	2.6VA AC coil
Max. Continuous	5.0W at 20℃ (68°F) ambient ,DC coil
Dissipation	7.0VA at 20℃ (68°F) ambient, AC coil
Temperature Rise	48°C(86°F) at nominal coil voltage, DC coil 68°C(122°F) at nominal coil voltage, AC coil
Temperature	Max. 155℃ (311°F)

Dimension

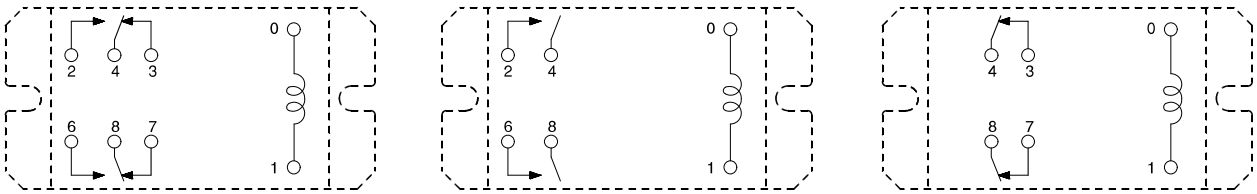
inch[mm]







Wiring Diagram



VIEWED TOWARD TERMINALS